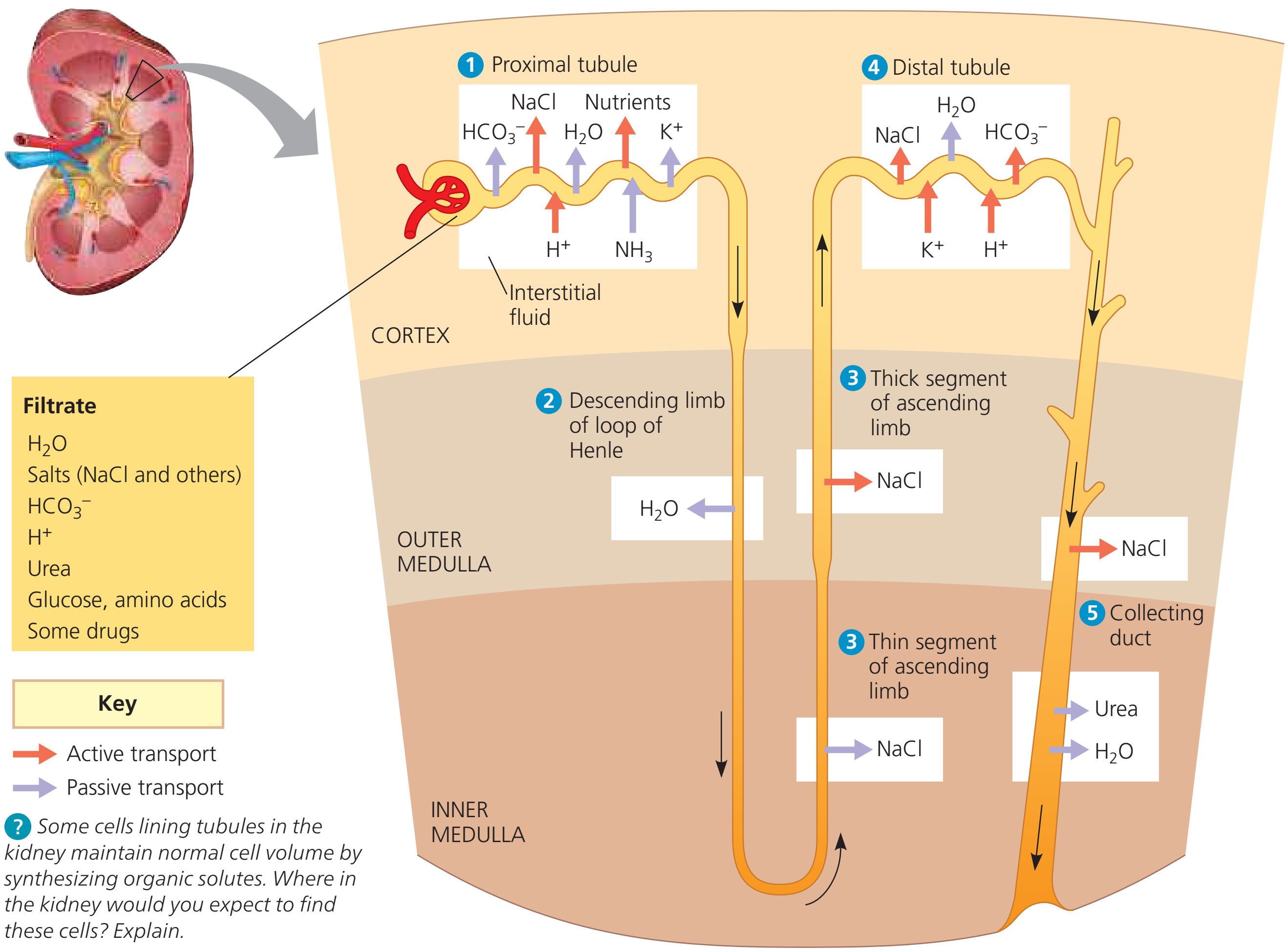
▼ Figure 44.13 The nephron and collecting duct: regional functions of the transport epithelium. The numbered regions in this diagram are keyed to the circled numbers in the text discussion of kidney function.



? *Some cells lining tubules in the kidney maintain normal cell volume by synthesizing organic solutes. Where in the kidney would you expect to find these cells? Explain.*

Mastering Biology Animation: Nephron Function

in transport epithelia as the filtrate moves through the kidney cortex and medulla **(Figure 44.13)**.

1 **Proximal tubule.** Reabsorption in the proximal tubule is critical for the recapture of ions, water, and valuable nutrients from the huge volume of initial filtrate. NaCl (salt) in the filtrate enters the cells of the transport epithelium by facilitated diffusion and cotransport mechanisms. There, Na^+ ions are transferred to the interstitial fluid by active transport (see Concept 7.4). This transfer of positive charge out of the tubule drives the passive transport of Cl^-.

As salt moves from the filtrate to the interstitial fluid, water follows by osmosis, reducing filtrate volume considerably. The salt and water that exit the filtrate diffuse from the interstitial fluid into the peritubular capillaries. Glucose, amino acids, potassium ions (K^+), and other essential substances are also actively or passively transported from the filtrate to the interstitial fluid and then into the peritubular capillaries.

Processing of filtrate in the proximal tubule helps maintain a relatively constant pH in body fluids. Cells of the transport epithelium secrete H^+ into the lumen of the tubule but also synthesize and secrete ammonia, which acts as a buffer to trap H^+ in the form of ammonium ions (NH_4^+). The more acidic the filtrate is, the more ammonia the cells produce and secrete, and a mammal's urine usually contains some ammonia from this source (even though most nitrogenous waste is excreted as urea). The proximal tubules also reabsorb about 90% of the buffer bicarbonate (HCO_3^-) from the filtrate, contributing further to pH balance in body fluids.

As the filtrate passes through the proximal tubule, materials to be excreted become concentrated. Many wastes leave the body fluids during the nonselective filtration process and remain in the filtrate while water and salts are reabsorbed. Urea, for example, is reabsorbed at a much lower rate than are salt and water. In addition, some materials are actively secreted into the filtrate from surrounding tissues. For example, drugs and toxins that have been processed in the liver pass from the peritubular capillaries into the interstitial fluid. These molecules are then actively secreted by the transport epithelium into the lumen of the proximal tubule.